A Multi-Agent Reflexion Framework with Two-Stage RAG

Tanay Kinariwala, Mahir Shah, Hetanshi Shah , Kailas K. Devadkar
Department of Computer Engineering, Sardar Patel Institute of Technology, Mumbai, India
Emails: {tanay.kinariwala23, amit.mahir23, hetanshi.shah23, kailas_devadkar}@spit.ac.in

Abstract - The integration of Large Language Models (LLMs) into automated educational assessment presents significant opportunities for scalable grading; however, zero-shot LLM approaches frequently suffer from hallucinated feedback, fabricated evidence, and inconsistent adherence to grading rubrics. To address these reliability issues, we introduce an automated paper-checking system that combines a Two-Stage Retrieval-Augmented Generation (RAG) pipeline with a Multi-Agent Reflexion framework. The system processes mixed-media student submissions using a hybrid native-OCR extraction module to ensure high text fidelity and grounds evaluations in verified source materials (e.g., course textbooks) through bi-encoder candidate retrieval followed by cross-encoder semantic re-ranking. The core grading mechanism is orchestrated by an Actor–Critic multi-agent loop, in which an Actor agent generates an initial rubric-based evaluation and a Critic agent audits it for factual consistency against retrieved evidence and detects unsupported or hallucinated claims. Evaluations that fail the audit are iteratively revised through a structured self-reflection step. Empirical results demonstrate that the proposed architecture reduces hallucinated grading outputs by x% and improves rubric adherence by x% compared to zero-shot LLM baselines, while generating interpretable, evidence-backed feedback suitable for high-stakes educational evaluation.

***Index Terms*—Automated Grading, Large Language Models (LLMs), Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Educational Technology, Fact Verification, Optical Character Recognition (OCR), Semantic Search.**

I. Introduction

The rapid advancement of Large Language Models (LLMs) has catalyzed significant interest in automated educational assessment, offering the potential to scale personalized feedback and reduce the administrative burden on educators. Traditional automated grading systems have historically relied on rigid keyword matching or regular expressions, which fail to capture the semantic nuances of student responses. While modern generative AI approaches address this by leveraging deep contextual understanding, deploying zero-shot or minimally prompted LLMs in high-stakes grading scenarios introduces critical vulnerabilities. Chief among these are the propensity for models to hallucinate factual information, fabricate supporting quotes from student texts, and deviate from strictly defined grading rubrics.

To bridge the gap between generative flexibility and rigorous academic assessment, we propose a novel, multi-agent automated paper-checking system designed to enforce factual grounding and strict rubric adherence. Our approach fundamentally departs from standard single-pass LLM wrappers by integrating a localized, verifiable knowledge base with a self-correcting agentic architecture.

To address the challenge of ingesting messy, mixed-media student submissions, our system implements a hybrid data extraction pipeline. It dynamically routes document pages through native PDF text extraction or optical character recognition (OCR) based on text density, ensuring high-fidelity digitization of both typed and scanned handwritten responses.

Crucially, to prevent the LLM from accepting plausible but factually incorrect student answers, we introduce a Two-Stage Retrieval-Augmented Generation (RAG) pipeline. This module grounds the grading process in authoritative course materials (e.g., digitized textbooks). By utilizing a fast bi-encoder for candidate chunk retrieval followed by a highly accurate cross-encoder for semantic re-ranking, the system provides the grading agents with precise ”truth contexts”.

The core evaluation is orchestrated through a Multi-Agent Reflexion Loop consisting of an ”Actor” and a ”Critic”. The Actor generates a preliminary grade based on the student’s text, the instructor’s rubric, and the retrieved truth context. Subsequently, the Critic acts as an independent auditor, verifying that the Actor did not ignore factual contradictions and that all cited evidence quotes are genuine. Evaluations that fail this audit trigger a reflexive regeneration, forcing the system to correct its hallucinations before finalizing the score.

The primary contributions of this paper are threefold:

1) **Hybrid Ingestion for Academic Submissions:** A robust PDF processing module combining PyMuPDF and PyTesseract to handle variable-quality student document scans seamlessly.
2) **Fact-Grounded RAG for Assessment:** The application of a dual-encoder RAG architecture to anchor LLM grading decisions in verified textbook knowledge, effectively mitigating domain-specific hallucinations.
3) **Actor-Critic Reflexion Framework:** A multi-agent evaluation loop that audits proposed grades for rubric compliance and factual integrity, providing a self-correcting mechanism for automated scoring.

The remainder of this paper is organized as follows: Section II reviews related work in LLM-based assessment and RAG

systems. Section III details the proposed system architecture, including the extraction, retrieval, and multi-agent grading components. Section IV presents our experimental methodology and evaluation metrics. Section V discusses the results and system performance, and Section VI concludes the paper with directions for future research.

II. RELATED WORK

The evolution of automated educational assessment has progressed from rigid keyword-matching systems to sophisticated reasoning-capable artificial intelligence frameworks. This section reviews prior work across four major directions: traditional NLP and machine learning-based grading systems, Large Language Models (LLMs) in educational assessment, Retrieval-Augmented Generation (RAG) for factual grounding, and advanced prompting and multi-agent reasoning frameworks.

A. Traditional Machine Learning and NLP-based Assessment

Early automated grading systems primarily relied on keyword matching, often failing to capture the semantic richness and contextual nuances of student responses. To overcome this limitation, Rokade *et al.* [1] proposed an NLP-based grading framework incorporating ontology-driven synonym expansion and contextual word matching to move beyond strict keyword dependence. Geigle *et al.* [2] framed automated grading of complex assignments as a supervised learning and ordinal regression task, demonstrating that active learning with human-in-the-loop strategies can significantly reduce manual grading effort while maintaining reliability.

Subsequent research increasingly adopted machine learning architectures. Xu [3] evaluated multiple ML classifiers—including Naïve Bayes, Support Vector Machines, and Random Forests—for grading written and programming assignments, reporting improved robustness over rule-based systems. Comprehensive surveys by Kumar *et al.* [4] and Wang *et al.* [5] further document how educational data mining and predictive analytics techniques have enhanced performance evaluation and academic outcome prediction.

Beyond objective assessments, automated grading has expanded to subjective and collaborative dimensions. An AI-driven framework for evaluating student collaboration [6] demonstrates how fuzzy logic and ML methods can quantify soft skills. More recently, Ayaan and Ng [7] proposed hybrid semantic evaluation models leveraging deep embeddings (e.g., Universal Sentence Encoder) alongside traditional similarity metrics such as cosine and Jaccard similarity, highlighting the maturation of standalone NLP-based grading systems.

B. Large Language Models in Educational Assessment

The introduction of the Transformer architecture by Vaswani *et al.* [8] marked a paradigm shift in natural language understanding, enabling the development of Large Language Models (LLMs) such as GPT-3 [9] and GPT-4 [10]. These models demonstrated unprecedented few-shot and zero-shot reasoning capabilities, making them attractive candidates for automated assessment tasks.

Xu *et al.* [11] provide a comprehensive survey of LLM applications in education, noting their transformative role in personalized tutoring, automated feedback, and scalable evaluation systems. Earlier discussions by Gardner *et al.* [12] critically examined the integration of AI in assessment, questioning its validity, reliability, and pedagogical implications. More recent analyses [13] suggest that AI-driven systems are actively reshaping evaluation paradigms across educational ecosystems.

Empirical evidence supports LLM-based grading potential. Uyar and Büyükahıska [14] evaluated ChatGPT as an Automated Essay Scoring (AES) tool and reported strong correlation with human raters under standardized rubrics (e.g., IELTS), though notable statistical discrepancies remained. Despite scalability and cost-reduction benefits, Wang *et al.* [5] caution against inherent risks, including bias, limited explainability, and hallucinated feedback—limitations that constrain standalone LLM deployment in high-stakes evaluation.

C. Grounding Evaluations with Retrieval-Augmented Generation (RAG)

To address hallucination and rubric misalignment in zero-shot LLM grading, researchers have proposed Retrieval-Augmented Generation (RAG) mechanisms. Lewis *et al.* [15] introduced RAG as a hybrid framework combining parametric knowledge within LLMs with non-parametric memory retrieved from dense vector indices. This approach enables models to ground their responses in external evidence.

Izacard and Grave [16] demonstrated that generative models significantly improve factual accuracy in open-domain question answering when augmented with retrieved passages. In educational assessment, RAG provides a mechanism to anchor grading decisions to verified instructional materials, rubrics, and course-specific references, thereby improving transparency and reliability.

D. Advanced Prompting and Multi-Agent Reflexion

While RAG enhances factual grounding, complex evaluation tasks demand structured reasoning. Wei *et al.* [17] introduced Chain-of-Thought (CoT) prompting, demonstrating that explicitly generating intermediate reasoning steps significantly improves logical and arithmetic problem-solving performance in LLMs.

Beyond prompting strategies, multi-agent conversational frameworks have emerged as a promising architectural paradigm. Wu *et al.* [18] proposed AutoGen, a framework enabling multiple specialized LLM agents (e.g., Actor and Critic) to collaborate iteratively on complex tasks. Extending this idea, Shinn *et al.* [19] introduced Reflexion, wherein agents maintain dynamic memory and generate self-reflective summaries to iteratively correct prior mistakes without weight updates.

E. Summary and Research Gap

Existing automated grading approaches range from standalone NLP similarity models [7] to direct LLM-based scoring systems [14]. However, zero-shot LLM grading remains vulnerable to hallucinations, rubric drift, and inconsistent reasoning. Conversely, advanced architectures such as RAG [15] and multi-agent Reflexion frameworks [19] have demonstrated substantial performance gains in knowledge-intensive and reasoning-heavy tasks but remain underexplored within rigorous academic evaluation contexts.

Therefore, a research gap exists at the intersection of semantically grounded grading and structured multi-agent reasoning. The proposed framework addresses this gap by integrating a two-stage RAG pipeline with an Actor-Critic multi-agent loop, enabling automated grading that is semantically aligned, factually grounded, and iteratively self-correcting.

III. Methodology

A. System Overview

This study proposes an automated grading framework that evaluates student responses against a trusted knowledge source using a retrieval-grounded large language model (LLM) pipeline. The system is designed to reduce hallucinations and improve grading reliability through structured evidence verification and multi-agent evaluation. The overall architecture consists of four primary components: Hybrid Heuristic-Based Document Parsing, Two-Stage Dense Retrieval with Cross-Encoder Re-Ranking, a Multi-Agent LLM Architecture utilizing a Reflexion/Critique Loop, and Structured Output with System Metrics.

The pipeline begins by ingesting student answer sheets in PDF format and converting them into machine-readable text using a hybrid extraction strategy. The extracted content is then evaluated against a curated knowledge base constructed from authoritative course material. To ensure factual grounding, the system employs a two-stage retrieval pipeline that first identifies candidate knowledge segments and subsequently re-ranks them using a cross-encoder model to produce a high-confidence context set. This retrieved context forms the factual basis used by the grading agents.

The evaluation stage is implemented using a role-based multi-agent architecture. An initial grading agent generates a rubric-aligned assessment of the student response, after which a secondary auditing agent verifies the factual consistency of the evaluation. If inconsistencies or low confidence are detected, a senior grading agent revises the assessment using critique feedback. This iterative Reflexion loop improves reliability and reduces unsupported reasoning.

Finally, the system produces a structured JSON output containing the assigned score, supporting evidence extracted from the knowledge base, and a reasoning trace for each evaluated question. Additional metadata such as processing latency and document extraction statistics are also recorded to support transparency and reproducibility of the grading process.

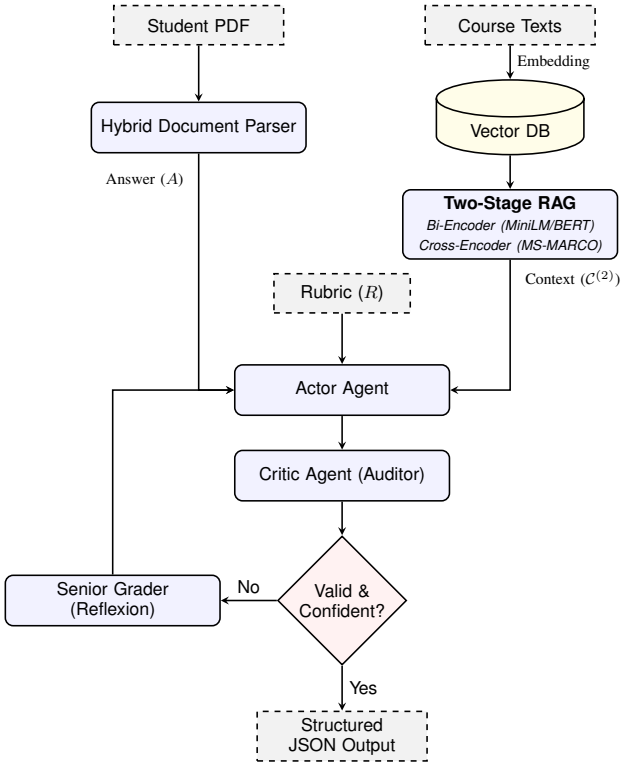

Fig. 1. End-to-end processing pipeline, showing vector embeddings, specific RAG models (MiniLM, MS-MARCO), and defined data flow (A, R, $\mathcal{C}^{(2)}$) into the Reflexion loop.

B. Hybrid Heuristic-Based Document Parsing

Student submissions often contain a mixture of digitally generated text and scanned handwritten or image-based content, which makes reliable text extraction challenging. To address this variability, the proposed system employs a **Hybrid Heuristic-Based Document Parsing** strategy designed to maximize extraction accuracy while maintaining processing efficiency.

Let a submitted document D be defined as a sequence of n pages such that:

$$D = \{p_1, p_2, \ldots, p_n\}$$

The pipeline begins by attempting native text extraction on each page p_i using the *PyMuPDF* library. This method is computationally efficient and preserves the original textual structure when the document contains embedded text. Let $T(p_i)$ represent the string of text extracted natively from page p_i.

However, many real-world student submissions consist of scanned pages where standard parsing fails. To systematically detect such cases, the system applies a heuristic evaluation on each page by measuring the character length of the extracted textual content, denoted as $|T(p_i)|$.

We define a minimum character count threshold $\tau = 50$. If

$$|T(p_i)| < \tau$$

the system classifies the page as likely image-based or poorly encoded.

In these situations, the pipeline automatically switches to an Optical Character Recognition (OCR) process using the *Tesseract* engine through the *pytesseract* interface. The page image is rendered and passed through the OCR model to recover textual information that cannot be obtained through standard PDF parsing. Let $\text{OCR}(p_i)$ represent the text recovered via this optical process.

The final text extraction function $E(p_i)$ for any given page is formally defined as a piecewise function:

$$E(p_i) = \begin{cases} T(p_i), & \text{if } |T(p_i)| \geq \tau \\ \text{OCR}(p_i), & \text{if } |T(p_i)| < \tau \end{cases}$$

This hybrid approach ensures robustness across diverse submission formats while minimizing unnecessary OCR computation for pages that already contain accessible text. Additionally, the system records the specific extraction mapping applied to each p_i, enabling later analysis of document quality, OCR utilization rates, and overall pipeline performance. This logging mechanism also supports reproducibility and provides useful metadata for evaluating the effectiveness of the document ingestion stage.

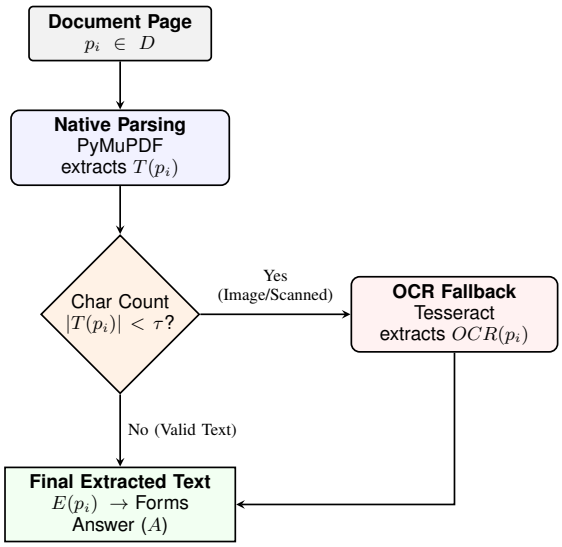

Fig. 2. The Hybrid Document Parsing strategy. Pages failing the minimum character threshold ($\tau = 50$) trigger the Tesseract OCR fallback to ensure high-fidelity text extraction (A).

C. Knowledge Base Construction

To ensure that the grading process remains grounded in authoritative course material, the system constructs a structured knowledge base derived from trusted academic sources such as textbooks, lecture notes, or instructor-provided reference documents. This knowledge base acts as the factual foundation for the retrieval-augmented grading pipeline and is generated during an offline preprocessing stage.

The source documents are first parsed and segmented into smaller textual units using a sliding-window chunking strategy. Let a parsed source document S be represented as a discrete sequence of m characters:

$$S = (c_1, c_2, \ldots, c_m)$$

We define a fixed chunk length (window size) $W = 1000$ and an overlap margin $O = 200$ characters between adjacent segments. The stride length, representing the step size between consecutive chunks, is therefore:

$$\Delta = W - O = 800$$

The i-th textual chunk C_i is extracted as the continuous subsequence:

$$C_i = (c_{i \cdot \Delta}, \ldots, c_{i \cdot \Delta + W - 1})$$

This overlap preserves semantic continuity and prevents important contextual information from being lost at chunk boundaries. The segmentation process produces a collection of context-preserving knowledge fragments suitable for semantic retrieval.

Following segmentation, each chunk is converted into a dense vector representation using a lightweight sentence-embedding model. Let

$$f_{\text{emb}} : \mathcal{X} \rightarrow \mathbb{R}^d$$

denote the embedding function that maps a discrete text sequence from the textual domain $\mathcal{X}$ to a d-dimensional continuous vector space. The embedding for the i-th chunk is computed as:

$$v_i = f_{\text{emb}}(C_i)$$

These embeddings capture the semantic meaning of the text, enabling similarity comparison between grading queries and the stored knowledge segments. The resulting vectors, along with their corresponding text content and metadata, are stored in a local JSON-based knowledge repository.

Formally, the generated knowledge base $\mathcal{KB}$ can be defined as the set of all chunk-vector pairs:

$$\mathcal{KB} = \{(C_i, v_i) \mid i \in \mathbb{N},\ 0 \leq i \cdot \Delta \leq m - W\}$$

This offline indexing process enables efficient retrieval during evaluation while avoiding repeated computation of embeddings. By structuring the reference material in this manner, the system ensures that subsequent grading decisions are grounded in verifiable information, forming the basis for the retrieval and verification stages described in the following sections.

D. Two-Stage Dense Retrieval with Cross-Encoder Re-Ranking

To ensure that the grading process is grounded in the most relevant reference material, the proposed system employs a two-stage dense retrieval pipeline that balances computational efficiency with semantic accuracy. Traditional single-stage retrieval methods often struggle to identify the most contextually appropriate passages, particularly when queries contain complex academic language or implicit concepts. The adopted approach mitigates this limitation by combining fast candidate retrieval with a more precise re-ranking stage.

In the first stage, a bi-encoder retrieval model is used to rapidly identify relevant knowledge segments from the pre-constructed knowledge base $\mathcal{KB}$. Let the evaluation query—derived from the grading rubric and question context—be denoted as q. Both the query q and the stored knowledge chunks are encoded into dense vector representations using the same sentence-embedding model f_{emb}, such that the query vector is:

$$v_q = f_{\text{emb}}(q)$$

Cosine similarity is then computed between the query vector v_q and all stored chunk embeddings $v_i \in \mathcal{KB}$ to produce an initial relevance ranking. The bi-encoder similarity score s_{bi} is formally defined as:

$$s_{\text{bi}}(q, C_i) = \frac{v_q \cdot v_i}{\|v_q\| \|v_i\|}$$

From this process, the system selects the candidate set $\mathcal{C}^{(1)}$ consisting of the top N candidate chunks (typically $N = 20$) that are most semantically aligned with the query:

$$\mathcal{C}^{(1)} = \underset{C_i \in \mathcal{KB}}{\arg\max}^{(N)} s_{\text{bi}}(q, C_i)$$

While bi-encoder retrieval is efficient, it evaluates the query and documents independently, which can limit its ability to capture deeper contextual relationships. To address this limitation, the second stage applies a cross-encoder re-ranking model that jointly processes the query q and each candidate passage $C \in \mathcal{C}^{(1)}$. This model performs a more detailed semantic comparison and assigns a refined relevance score s_{cross} to every candidate retrieved in the first stage.

Let the cross-encoder be defined as a function:

$$f_{\text{cross}} : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$$

The refined score is computed as:

$$s_{\text{cross}}(q, C) = f_{\text{cross}}(q, C)$$

The passages are then re-ordered according to these refined scores, and the top K segments (default value $K = 5$) are selected as the final Truth Context $\mathcal{C}^{(2)}$ used during grading:

$$\mathcal{C}^{(2)} = \underset{C \in \mathcal{C}^{(1)}}{\arg\max}^{(K)} s_{\text{cross}}(q, C)$$

By combining high-speed initial retrieval with more precise semantic evaluation, this two-stage architecture significantly improves the quality of contextual grounding while maintaining practical computational performance. The resulting context set forms the factual basis for the multi-agent evaluation process described in the following section.

E. Multi-Agent LLM Evaluation Architecture

To improve the reliability and transparency of automated grading, the proposed system employs a multi-agent large language model (LLM) architecture in which multiple specialized agents collaborate to produce and validate grading decisions. Unlike traditional single-model evaluation systems, this architecture distributes responsibilities across distinct roles, allowing independent verification and reducing the likelihood of unsupported conclusions.

The evaluation process begins with an **Actor** agent, which functions as the primary academic grader. Let the student's extracted response be denoted as A, the defined grading rubric as R, and the retrieved Truth Context generated by the retrieval pipeline as $\mathcal{C}^{(2)}$. We define the Actor agent as a generative function $\mathcal{M}_{\text{Actor}}$. Using these inputs, the Actor produces an initial assessment E_0:

$$E_0 = \langle m, r, e \rangle = \mathcal{M}_{\text{Actor}}(A, R, \mathcal{C}^{(2)})$$

where m represents the proposed numerical score, r is the explanatory reasoning trace, and e is the supporting evidence drawn from the retrieved material. The agent is explicitly instructed to prioritize factual consistency with the provided context and to penalize answers that contradict verified information.

Following the initial grading step, a second agent known as the **Critic** performs an independent audit of the Actor's output. The Critic examines whether the evaluation correctly reflects the rubric, verifies that the cited evidence actually exists within the retrieved context, and checks for logical inconsistencies or overlooked information.

Let the Critic agent be defined as a verification function:

$$\mathcal{M}_{\text{Critic}} : (\mathcal{X}, \mathcal{X}, \mathcal{X}, \mathcal{X}) \to (\{0, 1\}, [0, 1], \mathcal{X})$$

In addition to a textual critique t_c, the Critic produces a binary validity indicator $v \in \{0, 1\}$ and a continuous confidence score $c \in [0, 1]$ representing the reliability of the initial evaluation. The Critic output is defined as:

$$O_C = \langle v, c, t_c \rangle = \mathcal{M}_{\text{Critic}}(E_0, A, R, \mathcal{C}^{(2)})$$

This role-based architecture introduces separation between generation and verification, which improves robustness and helps detect hallucinated evidence or weak reasoning. By structuring the grading workflow as a collaborative interaction between specialized agents, the system encourages more consistent, transparent, and academically grounded evaluation outcomes. The interaction between these agents is further refined through an iterative critique and revision process described in the following section.

F. Reflexion-Based Critique Loop

To further improve grading reliability, the system incorporates a **Reflexion-based critique loop** that enables iterative self-correction of the evaluation process. Automated grading systems often produce incomplete or weakly justified assessments when relying on a single inference step. The proposed approach addresses this limitation by introducing a structured feedback mechanism in which evaluation outputs are reviewed and refined before finalization.

After the initial assessment E_0 is generated by the Actor agent and audited by the Critic, the system evaluates the Critic's response

$$O_C = \langle v, c, t_c \rangle$$

to determine whether the grading decision is sufficiently reliable. Specifically, the Critic provides both a validity flag v and a confidence score c that reflects how well the evaluation aligns with the rubric and retrieved factual context. We define a minimum confidence threshold $\theta = 0.85$.

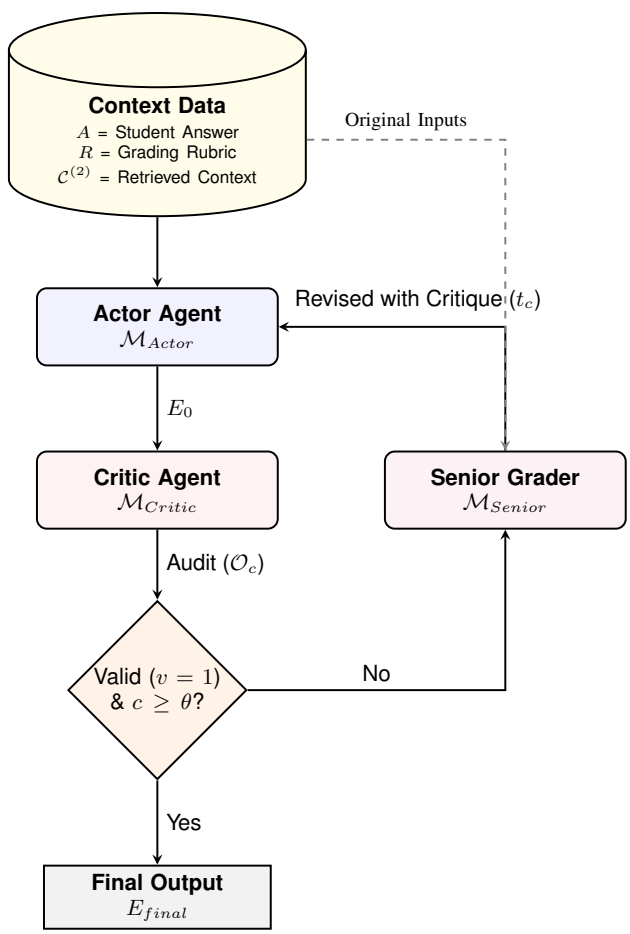

Fig. 3. The Multi-Agent Reflexion Loop. The context variables (A, R, $\mathcal{C}^{(2)}$) are routed to the Actor, while the Critic's audit triggers context-grounded revisions by the Senior Grader.

If the output is marked as invalid ($v = 0$) or if the confidence score falls below the predefined threshold ($c < \theta$), the system triggers a revision stage.

During this stage, a **Senior Grader** agent, denoted as the function $\mathcal{M}_{\text{Senior}}$, is invoked to regenerate the evaluation using the Critic's feedback as guidance. The Senior Grader receives the original inputs $(A, R, \mathcal{C}^{(2)})$ along with the initial evaluation E_0 and the critique notes t_c highlighting potential issues such as missing evidence, incorrect reasoning, or misalignment with the rubric.

By explicitly incorporating this feedback, the agent is able to revise the grading decision and produce a more accurate and justified assessment E_{revised}:

$$E_{\text{revised}} = \mathcal{M}_{\text{Senior}}(A, R, \mathcal{C}^{(2)}, E_0, O_C)$$

Formally, the final grading output E_{final} of the Reflexion loop is determined by the following piecewise function:

$$E_{\text{final}} = \begin{cases} E_0, & \text{if } v = 1 \text{ and } c \geq \theta \\ E_{\text{revised}}, & \text{otherwise} \end{cases}$$

This iterative critique mechanism encourages reflective reasoning within the multi-agent framework and reduces the likelihood of unsupported conclusions. The Reflexion loop therefore functions as a quality control layer, ensuring that the final grading output is consistent with both the retrieved knowledge context and the evaluation criteria defined by the rubric.

G. Evidence-Grounded Scoring

To ensure transparency and academic rigor in the grading process, the proposed framework adopts an **evidence-grounded scoring** mechanism that requires all evaluation decisions to be supported by verifiable context retrieved from the knowledge base. Rather than assigning scores based solely on generative reasoning, the system constrains the language model to justify each grading decision using explicit references to the retrieved Truth Context.

During evaluation, the grading agent analyzes the student's response in relation to the rubric criteria and the retrieved reference passages. Formally, let the overall grading rubric R be defined as a set of k distinct evaluation criteria:

$$R = \{r_1, r_2, \ldots, r_k\}$$

where each criterion r_j has a maximum assignable weight W_j.

For every rubric component r_j, the agent evaluates the conceptual alignment between the student's response A and the Truth Context $\mathcal{C}^{(2)}$ to determine whether it correctly reflects the verified concepts. We define an alignment function:

$$f_{\text{align}}(A, r_j, \mathcal{C}^{(2)}) \rightarrow \alpha_j$$

where $\alpha_j \in [0, 1]$ represents the degree of factual correctness and completeness. The awarded marks m_j for criterion r_j are therefore computed as:

$$m_j = \alpha_j \cdot W_j$$

This formulation ensures that marks are awarded proportionally to conceptual alignment and are reduced when important elements are missing, incomplete, or contradict the verified context.

To maintain interpretability, the system requires the grading agent to provide two structured explanations for each evaluated item. For each criterion r_j, the evaluation tuple is formally expressed as:

$$T_j = \langle m_j, \rho_j, \epsilon_j \rangle$$

where ρ_j represents the reasoning trace summarizing the logical process used to determine the assigned score, and $\epsilon_j \subseteq \mathcal{C}^{(2)}$ denotes an exact evidence quote extracted from the retrieved context that supports the evaluation.

The total score M for the response is computed as the summation over all rubric components:

$$M = \sum_{j=1}^{k} m_j$$

This dual-justification approach ensures that grading outcomes are not only explainable but also directly linked to authoritative source material.

By enforcing explicit evidence usage, the scoring mechanism reduces the likelihood of hallucinated justifications and encourages fact-consistent assessments. This design also enables instructors and researchers to audit grading decisions, making the evaluation pipeline more trustworthy and suitable for deployment in educational settings.

H. Structured Output Schema

To ensure interoperability, transparency, and ease of downstream analysis, the proposed system produces all grading results using a structured output schema. Rather than generating free-form textual feedback alone, the evaluation pipeline formats its final output as a standardized JSON object. Let the final structured output object be formally denoted as $\mathcal{O}$. This design allows the grading results to be easily integrated with front-end dashboards, analytics tools, and academic management systems.

The schema organizes the grading results into clearly defined fields that capture both the final scores and the reasoning behind them. Formally, the output $\mathcal{O}$ is defined as a tuple consisting of the grading evaluation array $\mathcal{G}$ and the system-level metadata $\mathcal{M}$ such that

$$\mathcal{O} = \langle \mathcal{G}, \mathcal{M} \rangle$$

Let

$$Q = \{q_1, q_2, \ldots, q_n\}$$

represent the set of evaluated questions. Each evaluated question $q_i \in Q$ is represented as an entry within the evaluation array $\mathcal{G}$.

Building on the scoring mechanism, each entry $E_i \in \mathcal{G}$ contains the assigned marks m_i, a reasoning trace ρ_i explaining the grading decision, and a supporting evidence quote ϵ_i extracted from the retrieved knowledge context. Therefore, the evaluation array is defined as

$$\mathcal{G} = [E_1, E_2, \ldots, E_n]$$

By enforcing this structure, the system ensures that every score is accompanied by a transparent justification grounded in verifiable source material.

In addition to grading information, the output also includes system-level metadata $\mathcal{M}$ generated during processing. This metadata records metrics such as total processing latency t_{total}, document parsing statistics, and the specific extraction methods $E(p_i)$ applied during the document ingestion stage (e.g., native parsing or OCR). Capturing these details enables reproducibility and provides insight into the internal behavior of the pipeline.

The use of a structured schema therefore serves two key purposes: it improves interpretability of automated grading decisions and enables systematic evaluation of the framework's performance. This standardized output format also facilitates future extensions of the system, including large-scale benchmarking, automated audits, and integration with educational platforms.

I. System Metrics and Logging

To evaluate the effectiveness and operational performance of the proposed grading framework, the system incorporates a dedicated metrics and logging module that records key information throughout the execution pipeline. These measurements provide insight into system behavior, enable reproducibility, and support empirical analysis in research settings.

During processing, the framework tracks multiple categories of metrics, including document ingestion statistics, retrieval behavior, agent evaluation outcomes, and overall processing latency. For the document parsing stage, the system records which extraction method $E(p_i)$ was applied to each page (native parsing or OCR). To quantify document quality and the frequency with which fallback OCR processing is required, we define the OCR utilization rate μ_{OCR} for an n-page document using an indicator function $\mathbb{I}$:

$$\mu_{\text{OCR}} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}\big(|T(p_i)| < \tau\big)$$

The retrieval stage logs the number of candidate chunks retrieved (N), re-ranking operations (K), and the final context size (total characters or tokens) used during grading. Monitoring these values allows researchers to evaluate the efficiency and effectiveness of the two-stage retrieval strategy.

Similarly, the multi-agent evaluation module records audit outcomes v, confidence scores c, and whether the Reflexion revision process was triggered. For a set of evaluated questions Q, the Reflexion trigger rate $\mu_{\text{Reflexion}}$ across the evaluation is formally expressed as:

$$\mu_{\text{Reflexion}} = \frac{1}{|Q|} \sum_{j=1}^{|Q|} \mathbb{I}\big(v_j = 0 \vee c_j < \theta\big)$$

In addition to component-level statistics, the system measures total end-to-end processing time for each evaluated submission. The total latency T_{total} can be decomposed into the sum of stage-wise execution times:

$$T_{\text{total}} = T_{\text{ingest}} + T_{\text{retrieve}} + T_{\text{evaluate}}$$

These latency measurements provide a practical assessment of scalability and usability in real educational environments.

All collected metrics are formally packaged as a metadata tuple

$$\mathcal{M} = \langle \mu_{\text{OCR}}, \mu_{\text{Reflexion}}, T_{\text{total}}, \dots \rangle$$

alongside the grading results within the structured output, enabling systematic analysis, benchmarking, and future optimization of the proposed framework.

J. End-to-End Pipeline

The complete grading workflow operates as a sequential pipeline that integrates document processing, retrieval, multi-agent evaluation, and structured reporting. The process begins when a student submission D is uploaded to the system in PDF format. The document is first processed using the hybrid heuristic-based parsing module, which attempts native text extraction and selectively applies OCR when necessary to recover unreadable or scanned content.

Formally, the extracted student response A is the concatenation of all parsed pages:

$$A = \bigoplus_{i=1}^{n} E(p_i)$$

Once the textual content is extracted, the system prepares the evaluation query q by combining the grading rubric R and question context. This combined query is defined as:

$$q = A \oplus R$$

The query q is passed to the two-stage dense retrieval module, which searches the pre-constructed knowledge base $\mathcal{KB}$ to identify the most relevant reference passages. The bi-encoder stage rapidly retrieves candidate segments $\mathcal{C}^{(1)}$, after which the cross-encoder re-ranking stage refines these results to produce a small set of high-confidence context passages referred to as the Truth Context $\mathcal{C}^{(2)}$.

The retrieved context $\mathcal{C}^{(2)}$, along with the student's response A and grading rubric R, is then provided to the multi-agent evaluation framework. The Actor agent generates an initial grading decision E_0, which is subsequently reviewed by the Critic agent for factual consistency and rubric compliance.

If the audit detects issues or produces a low confidence score ($c < \theta$), the Reflexion mechanism activates a Senior Grader agent to revise the evaluation using the critique feedback, ultimately yielding the final evaluation E_{final}.

After the evaluation stabilizes, the system compiles the results into a structured output format:

$$\mathcal{O} = \langle \mathcal{G}, \mathcal{M} \rangle$$

which includes scores, reasoning traces, supporting evidence, and system metrics. This final stage ensures that each grading decision is transparent, reproducible, and suitable for integration with external academic systems.

The entire end-to-end grading pipeline can therefore be summarized as a composite mapping:

$$\Phi : \mathcal{D} \times \mathcal{R} \to \mathcal{O}$$

which transforms a raw document submission and a rubric into a verifiable evaluation. Through this coordinated pipeline, the framework achieves reliable, evidence-grounded automated assessment while maintaining traceability across every stage of the evaluation process.

IV. Results

The proposed framework was evaluated on a dataset of 53 student assignments from a Distributed Computing File Systems course. Each assignment was graded manually by the instructor using a predefined rubric, and these scores were treated as ground truth.

Let y_i denote the actual (instructor-assigned) marks and $\hat{y}_i$ denote the predicted marks generated by the system for $N = 53$ samples.

A. Evaluation Metrics

To quantify grading performance, we compute standard regression metrics:

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{1}$$

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2} \tag{2}$$

Pearson Correlation Coefficient (r):

$$r = \frac{\sum_{i=1}^{N} (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{N} (y_i - \bar{y})^2 \sum_{i=1}^{N} (\hat{y}_i - \bar{\hat{y}})^2}} \tag{3}$$

B. Quantitative Results

The proposed system achieves the following performance:

- MAE = 2.30
- RMSE = 2.37
- Correlation (r) = 0.9897

C. Discussion

The low MAE and RMSE indicate that the predicted scores deviate by only 2–3 marks on average (out of 100), demonstrating high grading precision. The near-perfect correlation ($r \approx 0.99$) confirms strong agreement with instructor grading and consistent ranking of student performance.

These results validate the effectiveness of the proposed Two-Stage RAG and Multi-Agent Reflexion framework in producing accurate, reliable, and consistent grading outcomes.

V. Conclusion

This paper presented a reliable automated grading framework that integrates a Two-Stage Retrieval-Augmented Generation (RAG) pipeline with a Multi-Agent Reflexion architecture. By combining hybrid document parsing, retrieval-grounded evaluation, and an Actor–Critic verification loop, the system addresses key limitations of zero-shot LLM-based grading, particularly hallucinations and inconsistent rubric adherence.

Experimental results on 53 real-world assignments demonstrate strong alignment with instructor grading, achieving a low MAE of 2.30, RMSE of 2.37, and a high correlation of 0.9897. These findings validate that the proposed approach produces accurate, consistent, and interpretable grading outcomes.

The incorporation of retrieval grounding ensures factual correctness, while the Reflexion-based multi-agent loop improves reliability through iterative self-correction. Together, these components enable the system to generate evidence-backed evaluations suitable for academic assessment.

Future work will focus on scaling the framework to larger datasets, extending support for diverse subject domains, and optimizing latency for real-time deployment in educational environments.

References

[1] A. Rokade, B. Patil, S. Rajani, S. Revandkar and R. Shedge, "Automated Grading System Using Natural Language Processing," 2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT), Coimbatore, India, 2018, pp. 1123-1127, doi: 10.1109/ICICCT.2018.8473170.

[2] C. Geigle, C. Zhai, and D. Ferguson, "An Exploration of Automated Grading of Complex Assignments," in *Third (2016) ACM Conference on Learning @ Scale (L@S '16)*.

[3] D. C. F. Xu, "Design and Implementation of Machine Learning Algorithms in Automatic Grading of Students' Assignments," *Journal of Electrical Systems*, vol. 20-3s, pp. 899–919, 2024.

[4] C. V. A. Kumar, A. K. Eemani, G. C. Kalluri, and G. Rudra, "A survey on automated student evaluation and analysis using machine learning," *World Journal of Advanced Research and Reviews*, vol. 21, no. 2, pp. 2547–2554, 2024.

[5] J. Wang, Y. Hussain, X. Zhang, C. Mao, and Z. Wu, "AI-Based Automated Grading Systems: Opportunities, Challenges, and Future Directions," *Business and Social Sciences Proceedings Review (GEEA)*, 2025.

[6] B. B. Tomić, A. D. Kijevčanin, Z. V. Ševarac, and J. M. Jovanović, "An AI-based Approach for Grading Students' Collaboration," *IEEE Transactions on Learning Technologies*, vol. 16, no. 3, pp. 292–305, 2023.

[7] A. Ayaan and K.-W. Ng, "Automated grading using natural language processing and semantic analysis," *MethodsX*, vol. 14, p. 103395, 2025.

[8] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser and I. Polosukhin *et al.*, "Attention Is All You Need," in *31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.*.

[9] T. B. Brown *et al.*, "Language Models are Few-Shot Learners," arXiv preprint arXiv:2005.14165, 2020. Available: https://arxiv.org/abs/2005.14165.

[10] OpenAI, "GPT-4 Technical Report," arXiv preprint arXiv:2303.08774, 2023. Available: https://arxiv.org/abs/2303.08774.

[11] H. Xu, W. Gan, Z. Qi, J. Wu, and P. S. Yu, "Large Language Models for Education: A Survey," arXiv preprint arXiv:2405.13001, 2024. Available: https://arxiv.org/abs/2405.13001.

[12] J. Gardner, M. O'Leary, and L. Yuan, "Artificial intelligence in educational assessment: 'Breakthrough? Or buncombe and ballyhoo?,"' *Journal of Computer Assisted Learning*, vol. 37, no. 5, pp. 1207–1216, 2021.

[13] I. Saputra, A. Kurniawan, M. Yanita, E. Y. Putri, and M. Mahniza, "The Evolution of Educational Assessment: How Artificial Intelligence is Shaping the Trends and Future of Learning Evaluation," *Indonesian Journal of Computer Science*, vol. 13, no. 6, 2024.

[14] A. C. Uyar and D. Büyükahıska, "Artificial intelligence as an automated essay scoring tool: A focus on ChatGPT," *International Journal of Assessment Tools in Education*, vol. 12, no. 1, pp. 20–32, 2025.

[15] P. Lewis *et al.*, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[16] G. Izacard and E. Grave, "Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering," arXiv preprint arXiv:2007.01282, 2020. Available: https://arxiv.org/abs/2007.01282.

[17] J. Wei *et al.*, "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[18] Q. Wu *et al.*, "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation," arXiv preprint arXiv:2308.08155, 2023. Available: https://arxiv.org/abs/2308.08155.

[19] N. Shinn, F. Cassano, A. Gopinath, K. R. Narasimhan, and S. Yao, "Reflexion: an autonomous agent with dynamic memory and self-reflection," arXiv preprint arXiv:2303.11366, 2023. Available: https://arxiv.org/abs/2303.11366.